Braille mode

US keyboard coverage · UEB Grade 1 + optional Grade 2 · 8-dot Computer Braille

Reading the chord notation

A Braille cell is one simultaneous chord; adjacent cells are separate chords in order. An underlined cell means hold Braille Space with its dots. The underlined blank cell __ is Braille Space by itself (a literal host space).

Physical keys

Dots 1–3: `F D S` · dots 4–6: `J K L` · dot 7: `A` · dot 8: `;` · Braille Space: the key bound to `&bspc`.

UEB 1 · alphabet and numbers

Six-dot literary cells; ordinary US-layout host keystrokes

Lowercase letters

a	⠁	b	⠃
c	⠉	d	⠙
e	⠑	f	⠋
g	⠛	h	⠓
i	⠊	j	⠚
k	⠅	l	⠇
m	⠍	n	⠝
o	⠕	p	⠏
q	⠟	r	⠗
s	⠎	t	⠞
u	⠥	v	⠧
w	⠺	x	⠭
y	⠽	z	⠵

Digits

Start numeric mode with ⠼ then `a–j` cells mean `1–0`. A fresh numeric sign is needed after numeric mode ends.

1 ⠁ 2 ⠃ 3 ⠉ 4 ⠙ 5 ⠑
6 ⠋ 7 ⠛ 8 ⠓ 9 ⠊ 0 ⠚

Capitalisation and Grade 1 indicators

⠠	Capital next letter
⠠ ⠠	Capitals for the word
⠠ ⠠ ⠠	Capitals across a passage; ⠠ ⠠ ends it
⠰	Grade 1 symbol; ⠰ ⠰ word; ⠰ ⠰ ⠰ passage
⠰ ⠄	Grade 1 terminator

Indicators change reading context; they do not print as characters. Dot 6 is the capital prefix; dot 5-6 is the Grade 1 indicator.

UEB 1 · punctuation and symbols

All printable US keyboard characters have a literary sign

One-cell punctuation

,	⠂	;	⠆
:	⠒	!	⠖
apostrophe	⠄	hyphen -	⠤
period .	⠲	question ?	⠦

At a word start ⠦ opens a double quote; elsewhere it is `?`. To force `?` at word start, use ⠰⠦.

Multi-cell printable signs

` grave	⠈⠡	~ tilde	⠈⠔
!	⠖	@	⠈⠁
#	⠼	$	⠈⠎
%	⠨⠴	^	⠈⠢
&	⠯	*	⠈⠔
+	⠐⠖	=	⠐⠶
<	⠐⠣	>	⠐⠜
/	⠸⠌	_	⠨⠤
(	⠐⠣	)	⠐⠜
[	⠨⠣	]	⠨⠜
{	⠸⠣	}	⠸⠜
\	⠸⠡	|	⠸⠳

Quotation marks

Double opening	⠦ or ⠰⠦
Double closing	⠴ or ⠰⠴
Single opening / closing	⠠⠦ / ⠠⠴
Nondirectional quote	⠶ double · ⠄ single

All double forms emit "; all single forms emit ' on a US keyboard.

Example · type “Hello, world!”

UEB cells are sequential; the underlined blank Braille cell is the simultaneous Braille Space chord between words.

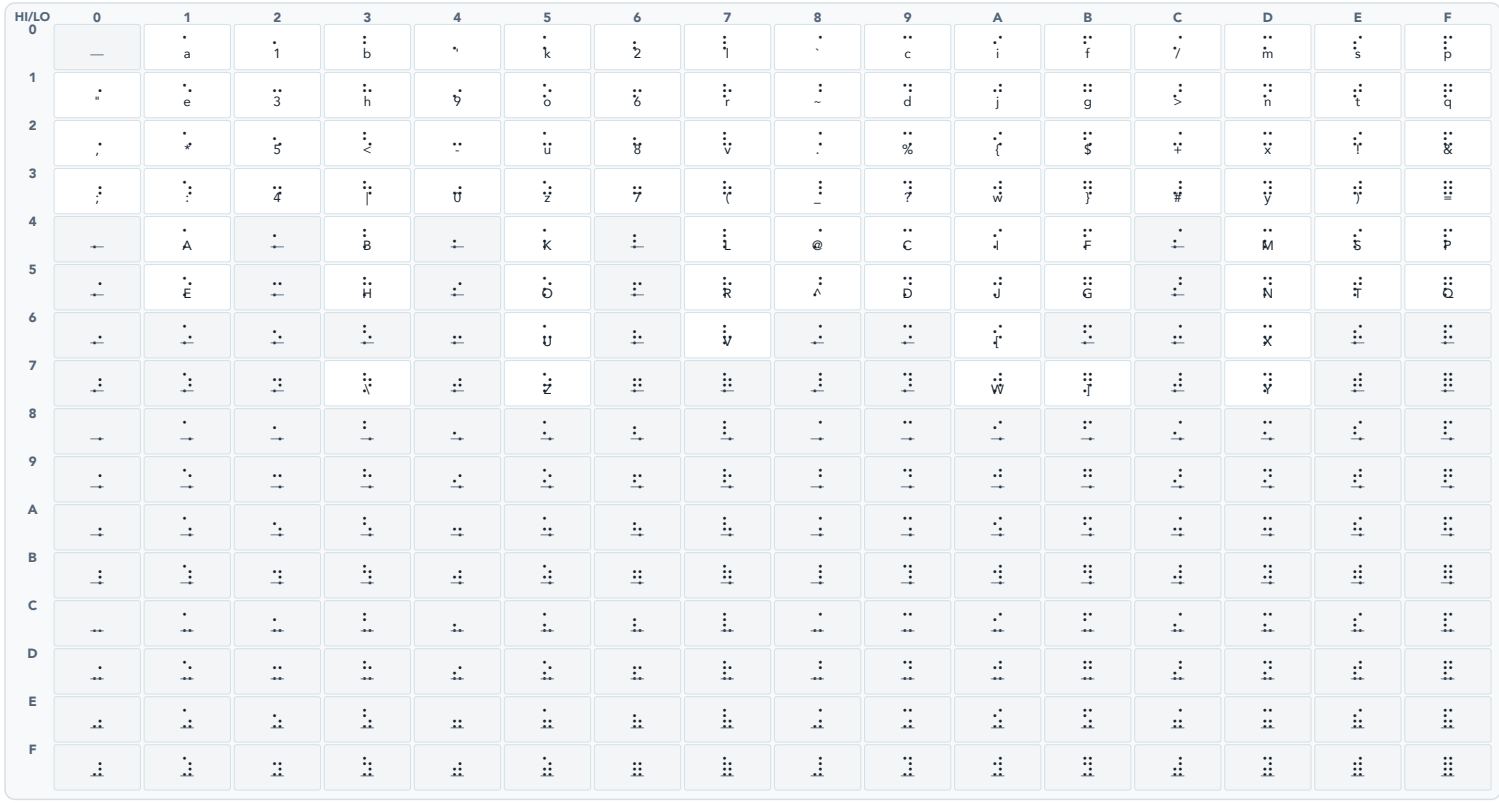
⠦ ⠓⠑⠇⠇⠕ __ ⠺⠕⠗⠇⠙ ⠖ ⠴

Computer Braille + actions

Dots 1–8 are independent; the ninth chord bit is Braille Space.

Computer8 · US English eight-dot table

Each cell below shows the Braille pattern and the printable ASCII character it emits; — is reserved. Row/column headers are hexadecimal mask nibbles.



Mask bit = dot: bit 0→1, … bit 7→8. The table contains all 256 patterns: 94 printable non-space ASCII characters map to dots; unmarked entries are reserved. Empty pattern cannot be pressed as a chord; Braille Space alone emits a literal space. Reserved patterns produce no host output and may be assigned by a user override. Source: liblouis `en-us-comp8.ctb` (US English 8-dot computer Braille), with its `loweredDigits6Dots.uti` and `latinLetterDef8Dots.uti` includes.

Optional UEB 2

Full ICEB reading inventory: 182 contractions

Alphabetic wordsigns — standing alone	23
Strong wordsigns — standing alone	6
Strong contractions	5
Strong groupsigns	12
Lower wordsigns	6
Lower groupsigns	10
Initial-letter contractions	33
Final-letter groupsigns	12
Shortforms	75

Wordsigns and shortforms depend on word boundaries; lower signs and initial/final contractions have positional rules. When a contraction condition fails, the UEB 1 letter/punctuation meaning applies. UEB 2 reads contractions; it does not choose where to write them.

Shared matching

Up to 8 chords per rule; longest valid sequence wins. A user override wins a same-length factory tie; up to 256 overrides persist.

Actions and commands

Factory command tables are mode-specific; commands can be extended with user rules

Keyboard / navigation actions

Left/right, Home/End, previous/next UI item; Tab, Return, Delete, Escape, arrows, Caps Lock; sticky Shift, Control, Option, Command; and F1–F12.

Host functions

Brightness down/up, Mission Control, Spotlight, previous track, play/pause, next track, mute, volume down/up, plus Globe as a host-dependent sticky modifier.

Command-chord conventions

An underlined dot pattern means hold those Braille keys with Braille Space. That extra chord bit can separate an action from a character; it is distinct from ordinary host Space.

Per-mode command bindings

Navigation, key, modifier, media/system and settings actions are planned. Their factory chords must be defined separately for each mode and checked against its character table—especially Computer8's full 256-pattern space.

Apple VoiceOver command assignments vary by display and can be customized in VoiceOver Utility; there is no single universal Apple mapping to copy.

Mode, input & settings

Persistent, on-device configuration

Character modes

`ueb1` is uncontracted UEB; `computer8` is 8-dot US ASCII; `ueb2` is an optional Grade 2 extension. Other named tables can be added without changing chord capture.

Chords and context

A chord contains every Braille key held from first press to last release—no timer. Braille Space is bit 9, separate from ordinary Space. Whitespace commits pending text; layer/mode changes and non-Braille input reset context.

Settings & limits

Display-guided settings capture/edit rules, select and persist a character mode, search compiled ZMK actions, and keep pending input visible. The settings controls stay fixed; user overrides do not change them.

`&kp` actions send normal host keystrokes, not Unicode Braille. Globe shortcuts are host-dependent; a true Mac Fn key, Touch ID/power, Dictation/Siri and Do Not Disturb are not emulated.

UEB source: ICEB, **The Rules of Unified English Braille**, 3rd ed. (2024), with the project's verified research tables. Apple's VoiceOver braille commands are assigned per display and may be customized; this guide describes planned module action coverage, not universal Apple default chords. User action overrides and the selected mode persist across reboot; factory rules are compiled.